Palm oil tank Retention & Melting

Info: 3x10 KL Tank responsible for long term ~~heat~~ storage and melting it to keep retention tank replenished. 1 x 5 KL Tank responsible for immediate storage of palm oil for use.

10 KL	5 KL
• Not insulated	Insulated
• Q_{batch}	$Q_{balance}$

Note: The tank specification is provided by optimising the manufacturing process for this project. Therefore, the sizing step is jumped for this worked example.

1) Batch target: 2 Ton palm oil

Batch time: 24 hrs, 20 hrs to be safe

Now,

$$Q_{s1} = 2000 \times 1.98 \times (36-25)$$
$$= 43560$$

$$Q_{s2(m)} = 2000 \times 187$$
$$= 374000$$

$$Q_{s3} = 2000 \times 2.15 \times (40-36)$$
$$= 17200$$

$$Q_T = 434730 + (0.4 \times 434730)$$
$$= 608622$$
$$\simeq \cancel{610\ KJ}\ \cancel{608622\ KJ}$$
$$\simeq 610000\ KJ$$

$$Q_{batch} = \frac{610000}{20 \times 3600} = 8.47\ KW$$

$$\simeq 9\ KW$$

2) Fluid property : maintained at 60°C (Water)

$$\rho = 983.2$$
$$C_p = 4185$$
$$\mu = 0.467 \times 10^{-3}$$
$$\nu = 0.475 \times 10^{-6}$$
$$K = 0.654$$
$$Pr = 2.99$$

From appendix B.

3) Tank geometry can be made from the help of ~~pa~~ handbook, here, for manufacturability and cost balace, measurements were provided by the optimizer.

$T_5 = 1.8 m (d)$ $T_{10} = 2.32 m (d)$
$2 m \; (h)$ $2.2 m \; (h)$

Note: T_5 has conical bottom but is of no use for us

~~4)~~
4) Pipe sizing : The hot water is to be supplied via CPVC pipe.

$$V_{max} = 1.5 \, ms^{-1}$$

We have 2 scenarios, 1" & 1½".

Smaller is better for cost optimisation

1"

$d_{out} = 0.03340$
$d_{in} = 0.02786$

$A_{in} = \pi d^2/4$
$= 6.096 \times 10^{-4}$
$C_{out} = 0.105$

1½"

$d_{out} = 0.04826$
$d_{in} = 0.04272$

$A_{in} = 1.433 \times 10^{-3}$

$C_{out} = 0.152$

Flow rate $(A_{in} \times V_{max})$.

Flow $= 9.144 \times 10^{-4} \, Kg s^{-1}$

$\simeq 3.292 \, Kg/hr.$

Flow $= 2.150 \times 10^{-3}$

$\simeq 7.7 \, Kg/hr.$

Engineering note : For coils, I have been cleared to use $1\frac{1}{2}$" SS pipe.

Pipe : $1\frac{1}{2}$"

when supplied via different pipes, velocity changes.

$$V_1 \times A_1 = V_2 \times A_2$$

$$V_2 = \frac{V_1 \times A_1}{A_2}$$

Via 1",

$V_2 = 0.688\ ms^{-1}$

via $1\frac{1}{2}$",

$V_2 = 1.5\ ms^{-1}$

For coil, if we deliver via 1", we need longer coil length and if we deliver via $1\frac{1}{2}$", the coil will be alright.

Now,

not reduced to fit more pipe.

$$d_{coil} = d_{tank} - 0.2 - d_{pipe}$$

$$= d_{tank} - 0.2$$

negligible

For T_5

$h = 1.9m$

$d_{coil} = 1.6m$

$p_{min} = 0.09652$

$N_{max} = h/p_{min} = 19$

$$l_{turn} = \sqrt{(\pi d_{coil})^2 + (p_{min})^2}$$

$$= 5.027$$

$$L_{max} = l_{turn} \times N_{max}$$

$$= 95.513m$$

For T_{10}

$h = 2.1m$

$d_{coil} = 2.1m$

$P_{min} = 0.09652$

$N_{max} = 21$

$$l_{turn} = \sqrt{(\pi . d_{coil})^2 + (P_{min})^2}$$

$$= 6.598$$

$$L_{max} = l_{turn} \times N_{max}$$

$$= 138.560m$$

Packing efficiency:

$\eta_f = 0.85$, $\eta_s = 0.93$, $\eta_w = 0.94$, $\eta_t = 0.94$, $\eta_a = 0.88$

$\eta_{pack} = 0.6145$

$l_{practical} = \eta_{pack} \times l_{max}$

$l_{prac} = 60.552\ m$ $\qquad$ $l_{prac} = 85.214$

η_{pack} was considered very conservatively,

$l_{prac} = 65\ m$ $\qquad$ $l_{prac} = 90\ m$

$A_{coil} = 9.88\ m^2$ $\qquad$ $A_{coil} = 13.68\ m^2$

5) Head loss

$$\# \ Re = \frac{\rho \times V \times d_{in}}{\mu}$$

$$Re = \frac{983.2 \times 1.5 \times 0.04272}{0.467 \times 10^{-3}}$$
$$= 128736.308$$

$$De = Re \times \sqrt{\frac{d_{in}}{d_{out}}}$$
$$= 19791.497$$

$De = 2822.829$

(both turbulent)

$$\# \ F_{straight} = (0.79 \ln(Re) - 1.64)^{-2}$$

$F(s) = 1.707 \times 10^{-2}$ $\qquad$ $F(S) = 1.707 \times 10^{-2}$

$$\# \ F(H) = F(s) \times \left[1 + 3.54 \sqrt{\frac{d_{in}}{D_{helix}}}\right]$$

$F(H) = 0.0483$ $\qquad$ $F(H) = 0.0443$

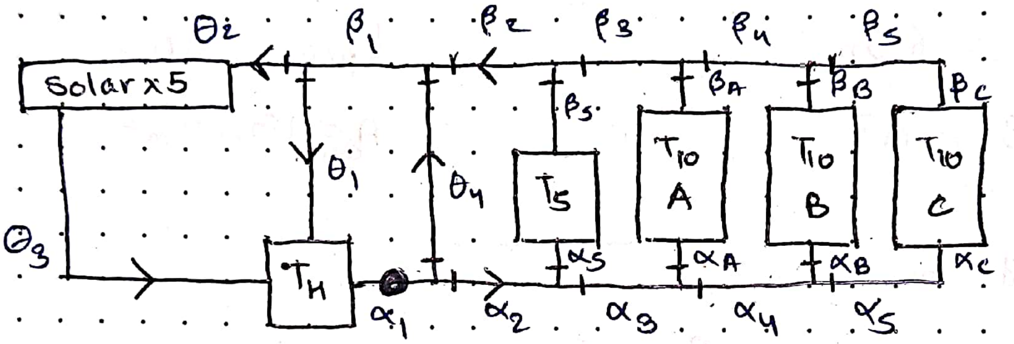
$$\# \ H_{coil} = F(H) \times \frac{L_{max}}{0.04272} \times \frac{1.5^2}{9.81 \times 2}$$

$H_{coil} = 8.427 m$ $H_{coil} = 10.703 m$

This is the max head for the coils if the coil is unchanged. For pump sizing, the distance from the pump factors a lot.

The head loss of pipes should be measured independently and differ from projects.

This is my proposal for the system design which has a passive solar heater for efficiency.



- . valves.

• . high pressure pump

> . Flow direction

here, Pump = 1 , Valves = 18

Possible loops : 1) Through solar
2) Bypass solar
3) Quick solar cycle

Each cycle has its own head depending on the length of pipe.

1) Through solar : $(\alpha + \beta) + \theta_2 + \theta_3$
↑
Selected Tank

2) Bypass solar : $(\alpha + \beta) + \theta_1$

3) Quick solar cycle = $\alpha_1 + \theta_4 + \beta_1 + \theta_2 + \theta_3$

Note : Solar will have its own independent head loss. There are 5 of them. although we can find that and add it to θ_2

Process control: Priority + Duty cycle.

→ $T_S > T_X$ (T_S is priority).

→ T_S operates on duty cycle. Duty cycle depends on the heat consumed by the T_S. When T_S is not in duty, T_X is priorised.

→ Through solar flow allows continious heat drain to the system and heat gain. It disturbs the passive solar though.

→ Bypass solar flow when energy generation through there is not good. OR when solar tank temperature reads low than threshold. The heat transmits through the thermal battery of the T_H.

→ Quick solar cycle allows for quick solar to thermal battery heat exchange as θ_g is governed by gravity. QSC allows the pump to promote it.

Note: Pressurising the tank of passive solar heater is a step into the dark - consequences are unknown to me, but I assume optimistic system behaviour.

6: Thermal performance (NTU - method, VCR framework)

Need to find U (considering $L_{practical}$)

Assuming T_S never solidifies while T_{10} solidifies.

For T_S, ($Pr = 2.99$, $f/8 = \cancel{2.134\times10^{-3}}$) $\rightarrow 0.00604$

$$Nu_{in} = f/8 \times 127.736.308 \times \cancel{2.718}\ 2.559 \times 1.075 \times 1.038$$

$$= \cancel{827.016}\ 2202.708 \qquad 33721.233$$

$$h_{in} = \frac{\cancel{827.016 \times 0.654}}{0.04292} = \cancel{12660.780}$$

$$\frac{1}{h_{in}} = \cancel{7.898\times10^{-5}}\ 2.965\times10^{-5}$$

$$R_{wall} = 0.04826 \times \frac{0.122}{2 \times K_{pipe}} \quad (SS = 13)$$

$$= 1.962 \times 10^{-4}$$

$R_{f_{in}}$ = (8.5, 1.5 ms^{-1}, soft water, low cycling)
$= 1.0 \times 10^{-4}$

$R_{f_{out}}$ = (Palm oil, 60°C, convection)
$= 7.0 \times 10^{-4}$

Now for h_{out}, No agitation (churchil - chu)

$\cancel{Nu_{out} = 0.6 +}$ $$Ra = \frac{9.81 \times 7.4 \times 10^{-4} \times 19 \times (0.04826)^3}{2.2 \times 10^{-5} \times \cancel{7.8 \times 10^{-8}}\ 0.028}$$

$= 9.034$

$P_r = 665$

$$Nu_{out} = \left(0.6 + \frac{0.559}{1.005}\right)^2$$

$= 1.337$

$$h_{out} = 1.337 \times \frac{0.152}{0.04826} = 4.211$$

$$\frac{1}{h_{out}} = \cancel{0.37}\ 0.237$$

$$\frac{1}{U} = 0.239 \quad \therefore U = 4.192$$

$C_r = 0,$

$$\varepsilon = 1 - e^{-NTU}$$

$$NTU = \frac{4.192 \times 9.88}{(9.114 \times 10^{-4} \times 983.2) \times 4185}$$

$= 0.01104$

$\varepsilon = 0.01098$, $C_{min} = 0.896 \times 4185$
$= 3750.130$

case A ($\Delta T = 24°C$) case B ($\Delta T = 20°C$)

$Q_{actual} = 993.75$ $Q_{actual} = 791.25$

XMD Page

Now, For the 10K Tanks,

For T_{10} ($Pr = 2.99$, $f/8 = 0.00554$)

$$Nu_{in} = \frac{f}{8} * 127736.306 * 2.574 \times 1.075 \times 1.030$$

$= 2016.933$

$$h_{in} = \frac{2016.933 \times 0.654}{0.04272} = 30877.203$$

$$\frac{1}{h_{in}} = 3.239 \times 10^{-5}$$

$R_{wall} = 1.962 \times 10^{-4}$ $R_{fout} = 7.0 \times 10^{-4}$

$R_{fin} = 1.0 \times 10^{-4}$ Now, h_{out}

U for melting, U for heating liquid

No; temp↑ even before all melts.

~~Need to find~~

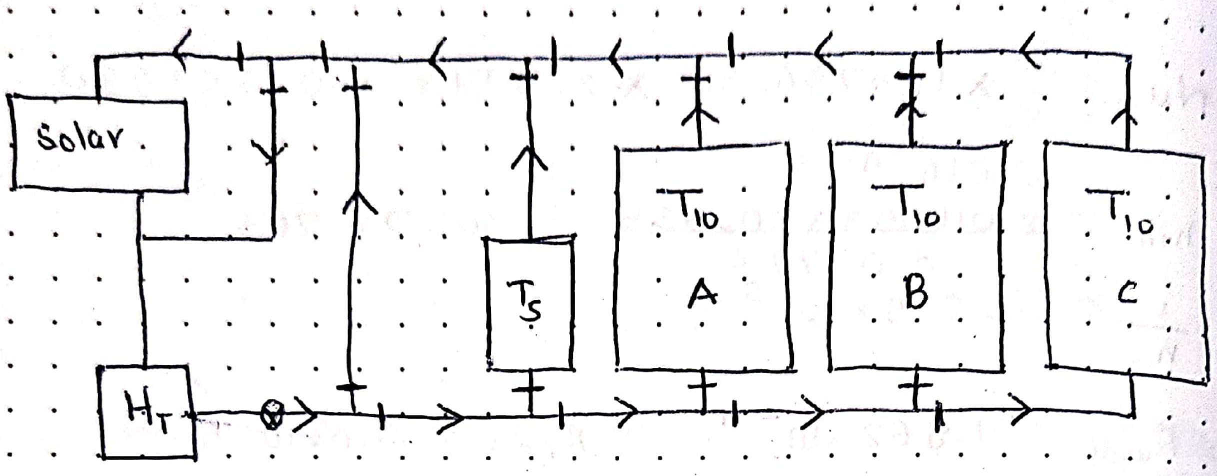
~~U melting~~

1). $U_{melting}$ 2) $\mathcal{E}_{solid}$ 3). $T_1 = E_1 / Q_{solid}$

4). $\mathcal{E}_{melting}$ 5) $Q_{melting}$ 6.) $T_2 = E_2 / Q_{meltin}$

$\pi d L * = 13.68$

$$L = \frac{13.68}{\pi \times 0.04826}$$



$9 \times 2 = 18$ values , 1 HPP

\> Flow direction

$\top$, $\bot$ values

⊗ High pressure pump

T_S temp	θ_S	$\theta_S < 40$	
T_S level		< 50%	
T_{10} batch release	check temp of T_{10} all, check volume of T_{10} all		target high volume one ↓ in emergency target high temp.
Solar recharge	solar temp	< 80°C	bypass
heat recharge	solar temp > 65 & H_T < 65		

T_{10} ? as long as T_S is not needed, give to T_{10} untill T_S > 90%

(T_S 90 ≥ TS ≥ 70)